

Waypoint Data Model

Team Nova | Tech Triathlon 2026 | Hackathon submission

Waypoint's four role workspaces share one set of operational records. This model connects outlet orders to depot plans, vehicle trips, loading evidence, delivery outcomes, and store receipts. It also records status history, exceptions, notifications, and offline synchronization.

The executable source of truth is `packages/database/prisma/schema.prisma`. This document consolidates the supplied implementation summary and earlier design document. Entity names follow the implementation summary; field types, nullability, foreign keys, and deletion policies must be read from the schema. The schema itself was not supplied for this review.

Network and identity

Entity	Purpose
Depot	Operating hub associated with outlets, vehicles, and service-da
Outlet	Brand location with district, dock, access, delivery window, add coordinates.
Vehicle	Depot fleet record with type, temperature capability, capacity, workshop state.
Account	Authenticated Store Manager, Dispatcher, Loader, or Driver, w depot scope.

Ordering and planning

Entity	Purpose
Order	Outlet request with service date, temperature, quantities, weig delivery window, and lifecycle status.
OrderLine	Product-level quantity, weight, and volume.
Plan	Versioned depot and service-date plan in draft, published, or s
Trip	Vehicle and Driver assignment, brand, district, and ordered sto
Allocation	Connects an order to a plan and trip, and records the stop sequ

Execution Evidence and Operational History

Execution and evidence

Entity	Purpose
LoadSession	Version-aware trip loading session with readiness evidence and confirmation state.
LoadLine	Planned and physically loaded quantities for an order line.
LoadIssue	Missing or damaged loading evidence, with an optional note and
DeliveryOutcomeRecord	Driver-recorded delivered or cannot-deliver result and receiver evidence.
ReceiptRecord	Store confirmation of expected and received quantities and the outcome.
ProofAsset	Extensible evidence metadata and storage reference. Its exact not specified in the supplied relationship diagram.

Audit exceptions and offline work

Entity	Purpose
OrderStatusEvent	Append-only explanation of changes to an order's lifecycle.
IssueCase	Shared exception record associated with an order.
IssueEvent	Lifecycle history of an issue case.
Notification	Account-targeted operational update linked to an entity.
AuditEvent	Actor, action, entity, time, and optional structured payload.
SyncOperation	Offline command identified by a unique operation ID, with service acknowledgement state.
SyncConflict	Conflict evidence associated with a synchronization operation.

Status history and audit records explain how the system reached its current state. They complement the current operational records rather than requiring every entity update to be represented only as a new row.

Entity Relationships

The relationships below reproduce the supplied core model in a readable reference. “Many” means zero or more unless a minimum is stated. They summarize intended cardinality; application rules may enforce requirements that foreign keys alone cannot express.

Network orders and planning

Parent	Related records
Depot	Many Outlets, Vehicles, and Plans.
Outlet	Many Accounts and Orders. Account outlet scope is optional in summary.
Account	Many Trips as Driver; many AuditEvents, Notifications, and Syn
Order	One or more OrderLines; many Allocations and OrderStatusEve
Plan	Many Trips and Allocations.
Vehicle	Many Trips.
Trip	One or more Allocations in the supplied diagram. Each Allocation order’s stop sequence.

Execution exceptions and synchronization

Parent	Related records
Order	Many IssueCases, LoadLines, and LoadIssues; at most one DeliveryOutcomeRecord and one ReceiptRecord.
Trip	At most one LoadSession; many LoadIssues.
LoadSession	Many LoadLines.
IssueCase	Many IssueEvents.
SyncOperation	At most one SyncConflict.

Reading the relationships

An Allocation joins an Order to its Plan and Trip. A LoadSession belongs to a Trip; its LoadLines capture the planned and loaded quantities. DeliveryOutcomeRecord and ReceiptRecord describe separate confirmations: the Driver’s result and the Store’s receipt.

The supplied diagram does not specify every evidence link, including ProofAsset associations and the direct OrderLine link for LoadLine. These should not be inferred as verified foreign keys without consulting the Prisma schema.

Integrity Rules and Workflow States

Uniqueness and record ownership

- Account email addresses are unique. Plan versions are unique within a depot and service date.
- An order occurs at most once in a plan. Stop sequences are unique within a trip. Vehicle and trip-number combinations are unique within a plan.
- Each trip has at most one LoadSession. Each order has at most one final DeliveryOutcomeRecord and one ReceiptRecord.
- Client-generated offline operation IDs are unique. LoadLines are unique by session and stable line key.
- Cascading deletion is limited to child records without an independent lifecycle. Exact relation policies are defined in Prisma.

Workflow state reference

Record	States and paths
Order	Main path: SUBMITTED → CONFIRMED → ALLOCATED → LOAD OUT_FOR_DELIVERY → DELIVERED. The supplied model also shows a CANNOT_DELIVER allocation and a CANNOT_DELIVER delivery outcome.
Plan	DRAFT → PUBLISHED → SUPERSEDED.
IssueCase	REPORTED → ACKNOWLEDGED → UNDER_REVIEW → RESOLVED. An IssueCase may be REOPENED and later resolved again.
SyncOperation	PENDING may become ACKNOWLEDGED, CONFLICT, or FAILED. A SyncOperation may be resolved; failed operations may be retried.
LoadSession	LOADING → CONFIRMED → REVERIFY_REQUIRED when re-verification is necessary.

These are the supplied state paths, not a complete transition matrix. “Resolved” and “retry” describe synchronization handling; they are not presented as additional enum values. CANNOT_DELIVER is described as an outcome rather than assumed to be an Order status.

Application enforcement

Application services enforce permitted transitions, actor scope, and prerequisites across related records. Unique keys and foreign keys provide structural integrity; workflow validation determines whether the signed-in role may perform an action and whether the related plan, load, or delivery state permits it.

Scope of this reference

Earlier design labels such as User, Load Record, and Sync Record are replaced here by the supplied implementation entity names. Deferral and Calendar Day are not listed as confirmed standalone entities because their schema definitions were not supplied. ARCHITECTURE.md explains the runtime components; the root README provides setup and the judge walkthrough.